

Benilde CLR Email Campaign Hub

Technical Documentation

Version 3.0

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Official Application URL:

<https://clr.benilde.edu.ph/emailcampaignhub>

Document Overview

This document serves as the official technical documentation for the **Benilde CLR Email Campaign Hub**, an internal web application developed to streamline the creation, review, organization, and management of CLR email marketing campaigns.

The documentation describes the application's architecture, technologies, workflows, data storage model, security implementation, and major system features to support future maintenance, enhancement, and knowledge transfer.

System Overview

The Benilde CLR Email Campaign Hub is a purpose-built, institutional email production system that solves the challenges of creating and managing branded Gmail campaigns for the Br. Fidelis Leddy Center for Learning Resources.

The Problem:

Before the Hub, CLR relied on plain-text Gmail drafts or Google Docs—a time-consuming process that often resulted in inconsistent branding, broken layouts, and campaigns that were hard to track or retrieve later. There was no formal approval trail, no image validation, and no way to ensure the CLR's visual identity was preserved across every email.

The Solution:

The Campaign Hub replaces manual HTML coding with a modular, drag-and-drop WYSIWYG builder that enforces the CLR brand guidelines. CLR can compose interactive, visually rich emails using pre-built templates and content blocks, preview the result in real-time, and generate Gmail-ready HTML with a single click.

Beyond creation, the Hub provides a full governance layer:

1. An approval workflow with audit trails (who approved what, and when).
2. A searchable archive that turns every campaign into a retrievable institutional asset.
3. A soft-delete and recovery system (30-day retention) to prevent accidental loss.
4. Automated maintenance (daily purges and image link checks) that keeps the system lean and healthy.

The result is a unified digital communication ecosystem where every email is on-brand, every decision is traceable, and every campaign is preserved for future reference.

The CLR Brand Experience: Why This Matters

The Hub is not just about efficiency, it is about **communication quality**.

In the digital age, a library's email is sometimes the first touchpoint for students, faculty, and partners. Plain-text emails communicate **disorganization**. Consistently branded, visually structured emails communicate **professionalism, clarity, and trust**.

With the Campaign Hub:

- **Non-technical staff** can now produce professional emails without needing HTML or graphic design skills.
- **Pubmats (publicity materials)** are rendered as responsive image grids with captions and click-through links, ensuring that the CLR's multimedia assets are displayed effectively.
- **Interactive elements** (CTA buttons with 5 style presets, clickable images, reference links) turn static emails into engaging calls-to-action.
- **The branded footer** (with social media icons, live chat, and contact hours) reinforces the CLR's identity in every single email.

The Transformation

What was once a manual, inconsistent process is now a **streamlined, brand-compliant, and auditable operation**—empowering the CLR to communicate with confidence, consistency, and impact.

Official Access URL

The Email Campaign Hub is officially accessed through the following permanent URL:

<https://clr.benilde.edu.ph/emailcampaignhub>

Rather than exposing the Google Apps Script deployment URL directly, the application uses a **LibGuides/Springshare Friendly URL** that redirects users to the current production deployment.

This approach allows future application deployments and version upgrades without changing the published application URL. Existing documentation, bookmarks, and training materials therefore remain valid even when the underlying deployment changes.

Architecture & Stack

Architecture Diagram

User -> CLR Friendly URL -> LibGuides Redirect -> Google Apps Script (Web App)

Technology Stack

Component	Technology
Frontend	Single-page HTML5 app with Tailwind CSS (CDN), vanilla JavaScript
Rich Text Editor	Quill 2.0
Screenshot Capture	html2canvas 1.4.1
Drag & Drop	SortableJS 1.15.2
Icons	Font Awesome 6.5.2
Backend	Google Apps Script (V8 runtime)
Metadata Storage	Google Sheets (18-column schema)
Payload Storage	Google Drive (JSON files per campaign)
Email Delivery	Gmail (MailApp.sendEmail)

Scope

The application supports:

- Email campaign creation with modular content blocks
 - Rich text editing and formatting
 - Image / pubmat management with bulk URL import
 - Gmail-ready HTML generation with a branded template
 - Campaign approval workflow with audit trail
 - Campaign archive management with search, filtering, and sorting
 - Soft deletion and recovery (30-day retention)
 - Test email delivery to authorized recipients
 - Image link validation and broken-link detection
 - Automated daily maintenance (purge expired items, link checking)
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User Authentication & Access Control

- **Google account-based authorization** via `Session.getEffectiveUser().getEmail()`

- **Whitelist of authorized users** (`ALLOWED_USERS` array in the backend code) – currently maintained by the development team
 - **Styled “Access Restricted” page** for unauthorized access attempts
 - Falls back to allowing access if no email is detected (web app runs as owner)
 - **OAuth scopes** used: `spreadsheets`, `drive`, `gmail.compose`, `script.send_mail`, `userinfo.email`
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Core Features

1. Campaign Studio (Draft Editor)

1.1 Quick-Start Templates

Nine pre-built templates provide a head start:

- Blank Canvas
- Event Announcement
- Workshop Invitation
- Resource Spotlight
- Service Update
- Library Hours Update
- Deadline Reminder
- Survey / Feedback
- Holiday Greeting

Each template populates the editor with a suggested structure (header, image, text, CTA) and placeholder content.

1.2 Campaign Metadata Fields

- **Target Send Date** – date picker (defaults to tomorrow)
- **Email Subject Line** – required, max 80 characters, soft warning at 70 chars with live counter and estimated read-time display
- **Target Audience** – free text (e.g., “For Students, For Faculty”)
- **Campaign Tags** – comma-separated free text

1.3 Approval Status & Revision

- Status dropdown: *Drafting* → *Pending Review* → *Approved for Sending* / *Revision Needed*
- **Revision Notes** textarea – appears only when *Revision Needed* is selected
- New campaigns submitted as *Pending Review* automatically trigger a review-nudge email to the deployment recipient

1.4 Storage Warning

A banner appears when the full campaign payload JSON exceeds **40 KB**, warning the user that the content will be stored in Drive (all content is stored in Drive regardless of size; this is a user-facing alert).

2. Email Section Builder

2.1 Section Types (4)

Type	Icon	Features
Header	fa-heading	Quill rich text editor; styled as a section divider with bottom border in preview
Text	fa-align-left	Quill rich text editor with full formatting toolbar
Image / Pubmat	fa-image	Compact inline editor with thumbnail preview, URL input, alt text, caption, click URL, reference URL, duplicate, and clear/remove
CTA Button	fa-link	Button text, URL, five style presets, alignment, and colour pickers

2.2 Rich Text Editor (Quill 2.0) Toolbar

- Bold, Italic, Underline
- Headings (H1, H2, H3, Normal)
- Text alignment (Left, Center, Right, Justify)
- Text colour and background colour
- Ordered and bulleted lists
- Insert link
- Clear formatting

2.3 CTA Button Style Presets

Preset	Style
Chunky	Heavy border, shadow, uppercase, bold
Classic	Rounded, solid background, normal weight
Pill	Fully rounded (999px), drop shadow
Outline	Transparent background, coloured border
Gradient	Gradient background (bg → text colour), shadow

2.4 Image Section (Compact Layout)

- **Thumbnail preview** (72×72 px, clickable)
- **Springshare asset URL input**
- **Alt text** field
- **Caption** field
- **Click URL** (wraps the image as a hyperlink)
- **Reference URL** field
- **Duplicate** button
- **Clear/Remove** image button

- **Image URL validation** status indicator (valid/invalid)

2.5 Bulk URL Paste Modal

- Text area for pasting multiple Springshare URLs (one per line)
- Each line creates a new Image section
- Confirmation shows the number of sections added

2.6 Section Management

- Add sections via type buttons (Header, Text, Image, CTA, Bulk URLs)
 - Drag-and-drop reordering (SortableJS, drag handle on each card)
 - Per-section actions: Move Up, Move Down, Duplicate, Remove
 - Staggered entry animation (**floatIn** with per-item delay)
 - Sections auto-scroll within a maximum height of 384 px
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3. Live Campaign Preview

- Real-time preview panel (sticky on desktop, scrolls independently)
 - Browser-style chrome with red/amber/green window dots
 - Displays:
 - Subject line
 - CLR sender avatar + name/email
 - Recipient ("To: Benildean Community")
 - Send date
 - Rendered email body with all sections composed in real time
 - **Mobile/Desktop toggle** – constrains preview to 375 px width for mobile simulation
 - **Estimated read time** (word count ÷ 200 WPM)
 - **Auto-layout for pubmats:**
 - Single image → centered, max 560 px
 - Multiple images → 2-column grid with captions and reference links
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4. Copy & Export Actions

4.1 Copy for Gmail

- Composes full Gmail-safe HTML (640 px table-based layout)
- Copies both **text/html** and **text/plain** to the clipboard via the **Clipboard API**
- Branded email shell:
 - Green top border
 - CLR header
 - Footer with "NEED HELP?" section
- Available from both the Draft Editor and the Preview Modal

4.2 Copy Preview (Screenshot)

- Captures the preview card as a **PNG** using **html2canvas** (2× scale for retina)

- Copies the image to the clipboard (falls back to download if **ClipboardItem** not supported)
- Visual flash animation during capture

4.3 Send Test Email

- Sends the complete campaign HTML to the configured deployment recipient via Gmail
- Available from the Preview Modal
- Uses the branded email shell wrapper

5. Archive Management

5.1 Archive Table

- **Server-side pagination** – configurable page size (default 10), navigable via Prev/Next and numbered page buttons
- **Status filter tabs**: All, Drafts, Pending, Approved, Revision
- **Search bar** – searches subject, author, status, and tags; clearable via × button
- **Sortable columns**: Send Date, Subject, Author, Status (click to toggle asc/desc)
- **Refresh button** for manual reload

5.2 Table Columns

Column	Details
Send Date	Formatted date from timestamp
Subject	Truncated with hover tooltip showing full text
Author	User email
Status	Coloured badge (Drafting/Pending/Approved/Revision) + broken-image indicator + approval info tooltip
Tags	Coloured pill badges per tag
Actions	Preview, Edit, Duplicate, Delete (trash) buttons

5.3 Archive Row Actions

- **Preview** (eye icon) – opens Preview Modal with full campaign rendering
- **Edit** (pen icon) – loads campaign into Draft Editor for modification
- **Duplicate** (copy icon) – clones to Draft Editor with "(Copy)" suffix, clears row ID
- **Delete** (trash icon) – soft-deletes with confirmation modal

5.4 Inline Status Change

- Status dropdown directly in archive table rows (no modal needed)
- Changing to *Approved* triggers the **Approval Audit Trail** modal
- Status change syncs to both Sheet metadata and Drive payload

- Visual feedback via toast notification
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6. Preview Modal

- Full-screen modal with backdrop blur
 - Displays:
 - Subject, author, target date, current status badge
 - Rendered email body
 - Reference URL section (with clickable link) if pubmat URLs exist
 - Inline status change dropdown with **Save** button
 - **Copy for Gmail** button
 - **Send Test Email** button
 - Closable via × button, Escape key, or click-outside
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7. Approval Workflow

7.1 Approval Audit Trail Modal

- Opens when setting status to *Approved for Sending*
- Preset approvers: Sir Michael, Miss Vanessa, Miss Anna, Sir Winston
- Custom approver option with text input
- Saves **approvedBy** (name) and **approvedAt** (timestamp) to both Sheet and Drive payload

7.2 Approval Info Display

- In the archive table: "By [Approver Name]" text with hover tooltip showing full timestamp
 - Auto-cleared when status changes away from *Approved*
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8. Trash / Soft-Delete System

8.1 Soft Delete

- Campaigns moved to trash have **IsDeleted=true**, **DeletedAt=timestamp**, **DeletedBy=user email**
- Hidden from all archive views but preserved in Sheet and Drive
- **30-day retention window** before auto-purge

8.2 Trash Tab

- Dedicated table view with columns:
 - Deleted At
 - Subject
 - Deleted By
 - Retention countdown
 - Actions
- **Retention** column shows "EXPIRED" (red) when past 30 days, or "X days left" (amber)

- **Restore** button – clears soft-delete flags, campaign reappears in archives
- **Delete Forever** button – permanently deletes row and Drive payload (confirmation required)
- **Purge Expired** – deletes all items past 30-day retention
- **Purge All** – deletes everything in trash regardless of age

8.3 Confirmation Modal

- Reusable custom modal dialog (replaces native `confirm()`)
- Configurable: eyebrow text, title, message, confirm button text, confirm button style (colour classes)
- Supports Escape key and click-outside to cancel

9. Broken Image Link Detection

- **Server-side URL validation** for all image sections (checks http/https protocol)
- **"broken" badge** visible in the archive table when image URLs are unreachable
- **Manual "Check Now" button** to run validation on demand
- Results stored in Sheet columns: **Image URL Status** (ok/broken/unchecked), **Image URL Checked At** (timestamp)

Automated Processes

Daily Cleanup Trigger

- Auto-purges expired soft-deleted campaigns (past 30-day retention)
- Configurable hour (default 2:00 AM Asia/Manila)
- Enable/disable via server functions
- Trigger status check available

Daily Broken-Link Checker Trigger

- Auto-validates all Springshare image URLs across all campaigns
- Configurable hour (default 3:00 AM Asia/Manila)
- Enable/disable via server functions
- Results logged to Script Properties and Logger

Data Storage Architecture

Google Sheets (18 Columns)

#	Column	Purpose
0	Timestamp	Target send date
1	Last Modified	Auto-updated on any change
2	Author	User email

#	Column	Purpose
3	Status	Drafting/Pending/Approved/Revision/Archived
4	Subject	Email subject line
5	Target Audience	e.g., "For Students"
6	Campaign Tags	Comma-separated
7	Approved By	Approver name
8	Approved At	Approval timestamp
9	Revision Notes	Feedback for revisions
10	Pubmat URLs	Legacy/compatibility field
11	Storage Mode	"Springshare" or legacy "Drive"
12	Payload File ID	Drive file ID of campaign JSON
13	IsDeleted	Soft-delete flag (Boolean)
14	DeletedAt	Soft-delete timestamp
15	DeletedBy	User who deleted
16	Image URL Status	ok / broken / unchecked
17	Image URL Checked At	Last validation timestamp

Google Drive Payloads

- Full campaign JSON stored in a dedicated folder (*Benilde CLR Campaign Payloads*)
- Each file named: `[subject]_[timestamp].json`
- Loaded **on-demand** only (never during archive listing – lazy loading)
- Files shared with "Anyone with the link can view"
- Folder ID cached in Script Properties

Caching

- Campaign tags cached for 10 minutes (Script Properties)
- Tag cache busted on every save

Email Composition & Rendering

Gmail-Safe HTML

- 640px max-width table-based layout
- Inline CSS only (no external stylesheets)
- Branded shell:
 - Green 6px top border
 - "Benilde CLR" header

- White content area

Rich Text → Email Conversion

- Strips Quill CSS classes, adds inline font-family/size/color/margin styles
- H1 → 26 px, H2 → 22 px, H3 → 18 px (all Arial, bold)
- Paragraphs: 14 px, 1.7 line-height
- Links: Benilde green (#006b3f), underlined, bold
- Text alignment preserved (center/right/justify)
- All **on*** event attributes stripped (security)

Branded Email Footer

- **"NEED HELP?"** hook
- Available hours: Mon–Sat, 8 AM – 5 PM
- **"Email Us"** button (<mailto:clr@benilde.edu.ph>)
- **"Live Chat"** button (links to clr-answers.benilde.edu.ph)
- **Website link** (clr.benilde.edu.ph)
- **"Connect with us"** section with social media icon links (FB, IG, X, YT, TikTok)
- Separator lines between sections

Pubmat Layout Logic

- **Single image** → centered, max-height 600 px (preview) / 400 px (email), full width
- **Multiple images** → 2-column grid, max-height 400 px (preview) / 300 px (email), responsive
- Captions and reference links rendered below each image
- Click URLs wrap images as hyperlinks

Security Measures

- **HTML Sanitization** (DOMParser-based) – removes `<script>`, `<style>`, `<iframe>`, `<object>`, `<embed>`, and all **on*** event handler attributes
 - **URL Normalization** – only `http://` and `https://` protocols allowed; all others rejected
 - **Output Escaping** – all user-facing dynamic text runs through `escapeHtml()` / `escapeAttr()`
 - **Server Authorisation** – `assertAllowedUser_()` called at the top of every server function
 - **Drive Payload Isolation** – campaign JSON files in a dedicated folder, shared view-only
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UI/UX Design System

- **Neobrutalist design language:** thick 2 px borders, heavy box-shadows, bold typography
- **Custom colour palette:**
 - Green (#006b3f), dark green (#02472c), mint (#e8f4dc)
 - Lime (#b7df52), yellow (#ffe01b), coral (#ff684f), sky (#bfe3ff)
 - Warm paper (#fffdf6), warm deep (#f6eddc)
 - Ink (#17231d)
- **Hover animations:** buttons translate up-left on hover, active translates down-right (3D button effect)

- **Entry animations:** staggered `floatIn` for sections
- **Table row hover:** slight translate-right + background highlight
- **Status badges:** colour-coded with distinct backgrounds (gray/yellow/green/red/purple)
- **Status select dropdowns:** coloured chevrons matching status colour
- **Custom scrollbar:** thin (6px), rounded track and thumb
- **Responsive:** mobile-friendly pubmat grid (1→2 columns), compact image sections
- **Header grid accent:** SVG mask with radial gradient overlay

Keyboard Shortcuts

Shortcut	Action
<code>Ctrl/Cmd + K</code>	Create new campaign
<code>Ctrl/Cmd + F</code>	Focus archive search bar
<code>Ctrl/Cmd + S</code>	Submit/save draft
<code>Escape</code>	Close modals (preview, approval, confirm), close toast

Health Check & Error Handling

- **Server health endpoint** (`checkServerHealth`) – returns status, user email, sheet row count, timestamp
 - **Client-side library check** – verifies Quill, html2canvas, SortableJS loaded; displays startup warning banner if missing
 - **Global error handler** (`window.onerror`) – logs to console
 - **Unhandled promise rejection handler** (`window.onunhandledrejection`) – logs to console
 - **Graceful server errors** – all server functions wrapped in `try/catch`, return structured error messages to client
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Bulk Operations

- **Bulk Archive Past Campaigns** (`bulkArchivePastCampaigns`) – auto-marks campaigns with past target dates as *Archived*
 - **Bulk URL Paste** – creates multiple image sections at once from pasted URLs
 - **Bulk Purge** – “Purge All” and “Purge Expired” in trash
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Future Enhancements

Potential future improvements to enhance the Email Campaign Hub include:

- **Campaign analytics dashboard** – track open rates and click-throughs
- **Email engagement reporting** – visualise campaign performance
- **Multi-level approval workflow** – support for multiple approvers
- **Scheduled campaign publishing** – send emails at designated times
- **Version history** – track changes and revert to previous versions

- **Campaign duplication templates** – save common structures as templates
- **AI-assisted subject line suggestions** – improve open rates
- **Direct Gmail Draft integration** – create Gmail drafts directly

Version History

Version	Date	Description
1.0	2026-06-25	Initial release – basic campaign builder
2.0	2026-07-10	UI redesign, archive system, improved preview, hyperlinked images, new approval trail flow, multi-pubmat grid rendering, and CTA button builder & spacing
3.0	2026-08-02	Approval workflow, soft-delete, preview improvements, payload storage, automated maintenance

Appendix

OAuth Scopes Used

- <https://www.googleapis.com/auth/spreadsheets>
- <https://www.googleapis.com/auth/drive>
- <https://www.googleapis.com/auth/gmail.compose>
- https://www.googleapis.com/auth/script.send_mail
- <https://www.googleapis.com/auth/userinfo.email>

Google Services

- **Google Sheets** – metadata storage
- **Google Drive** – campaign payload storage (JSON files)
- **Gmail** – email delivery (test emails and review nudges)
- **Google Apps Script** – backend logic and web app hosting

Libraries & CDN Resources

- **Tailwind CSS** (CDN)
- **Quill 2.0** (rich text editor)
- **html2canvas 1.4.1** (screenshot capture)
- **SortableJS 1.15.2** (drag-and-drop)
- **Font Awesome 6.5.2** (icons)

Folder Structure (Google Drive)

- Root folder: *Benilde Campaign Payloads*
- Each campaign stored as a JSON file with naming pattern: `[subject]_[timestamp].json`
- Folder ID cached in Script Properties for efficient access

Spreadsheet Schema

See [Data Storage Architecture](#) for the full 18-column schema.

Scheduled Triggers

Trigger Name	Function	Default Hour	Purpose
Daily Cleanup	<code>scheduledPurgeExpired_</code>	2:00 AM	Purge expired soft-deleted campaigns
Broken-Link Checker	<code>scheduledImageLinkCheck_</code>	3:00 AM	Validate all image URLs

Triggers can be enabled/disabled and the hour can be adjusted via the server functions (`setupDailyCleanupTrigger(hour)`, `setupImageLinkChecker(hour)`).

Conclusion

The **Benilde CLR Email Campaign Hub** centralises the CLR’s email marketing workflow into a single, maintainable platform. By combining campaign creation, approval management, automated storage, Gmail-ready email generation, and long-term archival within one application, the system reduces manual effort while improving consistency, traceability, and operational efficiency.

The application’s modular architecture, Google Workspace integration, and permanent access URL enable future enhancements and deployment updates without disrupting end users, making it a scalable solution for the CLR’s evolving digital communication needs.